

■ SLO-2 Department-Level Intervention: Concept Development

SLO-2: *An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.*

This rubric captures the department-level intervention for **Concept Development (Apply → Create)** in **Senior Design I (capstone)**.

Student Struggles (Summary)

Students are strong at **Evaluate** but show persistent weaknesses at **Apply** and high variability at **Create**. In Senior Design I, teams often struggle to articulate clear problem statements and to move from conceptual ideas to embodiment/specifications. Many need more support with problem formulation and constraint analysis. Performance dipped below 70% in F21 and F24 during open-ended tasks, before recovering in Sp25.

Overlay Rubric: Quick Steps (SLO-2 – Concept Development)

- **Grade as usual.** Keep using the existing SD-1 grading sheet (no new forms or extra scoring).
- **Focus on the Concept Development criterion.** This is already part of SD-1 Mid-Term Poster and related deliverables.
- **Apply the intervention lens while grading:** monitor whether teams (a) generate ≥ 6 design concepts, (b) use ≥ 1 structured creativity method (e.g., morphological chart, 6-3-5, SCAMPER), and (c) document feasibility and linkage to requirements/constraints.
- **Your task:** continue grading Concept Development exactly as you do now — just keep the intervention criteria in mind.

Performance Indicator (PI)	Level	Senior Design I (Capstone) Expectation	Bloom's Link
PI: Concept Development	Novice	≤ 3 low-variation ideas; no structured creativity method; ideas not tied to requirements/constraints; documentation minimal.	Apply
	Apprentice	4–5 ideas with partial/fragmented use of a method (e.g., incomplete morphological chart); weak linkage to key requirements; limited feasibility notes.	Apply
	Practitioner	≥ 6 diverse concepts using ≥ 1 structured method; each concept mapped to requirements/constraints; feasibility comments included; documentation clear.	Apply → Create
	Expert	Highly diverse concept set exploring different architectures/modalities; explicit rationale for diversity; professional, well-structured documentation (tiles/sketches/briefs) ready for screening and decision-making.	Create